



Integral Equations

Prepared by

Dr. Gamal Ahmed Mosa

Mathematics Department - Faculty of Science,
Benha University - Egypt

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1. Introduction

It is important to note that integral equations arise in engineering, physics, chemistry, and biological problems. Many initial and boundary value problems associated with the ordinary and partial differential equations can be cast into the integral equations of Volterra and Fredholm types, respectively. Engineering problems can be mathematically described by differential equations model. Similarly, problems arising in electric circuits, chemical kinetics, and transfer of heat in a medium can all be represented mathematically as differential equations. These differential equations can be transformed to the equivalent integral equations of Volterra and Fredholm types. On the other hand, there are many physical problems that are governed by the integral equations and these equations can be easily transformed to the differential equations.

The purpose of this review article is to clarify the importance of integral equations in natural and physical applications by providing the basic concepts of integral equations and the relations between differential equations with integral equations, moreover, how to convert between them and then focus on some different numerically solutions methods of integral equations

later in next sections we present some points to study as a future work.

2. Integral Equations

2.1 Integral equations

Integral equations (IEs) exist in nature and appear in various fields of science and engineering [10, 8, 83]. Accordingly, there are many computational approaches to approximate the solutions of IEs [26, 45, 46, 65]. The IE is an equation in which the unknown function $\phi(x)$ appears inside an integration. The most standard type of integral equation is:

$$r(x)\phi(x) = f(x) + \lambda \int_{\alpha(x)}^{\beta(x)} K(x, y)\phi(y)dy, \quad (2.1)$$

where

1. $r(x)$ is a function defining the kind of the IE,
 - i. if $r(x) = 0$, then the IE is of the first kind,
 - ii. if $r(x) = \mu \neq 0$, μ is constant, then the IE is of the second kind,
2. the kernel $K(x, y)$ is known function,
3. the function $f(x)$ is called free term of the IE,
 - i. If $f(x) = 0$ then equation (2.1) is called homogenous integral equation,
 - ii. If $f(x) \neq 0$ then equation (2.1) is called inhomogeneous integral equation.
4. the unknown function $\phi(x)$ appears inside and outside the integration that shall be determined and the linearity of the IE depends on its power.
5. λ is called the IE parameter.

2.2 Classification of integral equations

The linearity of the IE depends on the power of the function $\phi(x)$ as we have seen in the ordinary and partial differential equations. In this text, we shall distinguish three major types of integral equations. In particular, these types are given below:

1. Fredholm integral equations

$$r(x)\phi(x) = f(x) + \lambda \int_a^b K(x, y)\mathcal{N}_0(\phi(y))dy. \quad (2.2)$$

2. Volterra integral equations

$$r(x)\phi(x) = f(x) + \lambda \int_a^x K(x, y)\mathcal{N}_0(\phi(y))dy. \quad (2.3)$$

3. Mixed integral equations

$$r(x)\phi(x, t) = f(x, t) + \lambda \int_0^t \int_a^b F(t, \tau)K(x, y)\mathcal{N}_1(\phi(y, \tau))dyd\tau. \quad (2.4)$$

Or

$$\begin{aligned} r(x)\phi(x, t) = & f(x, t) + \lambda \int_a^b K(x, y)\mathcal{N}_2(\phi(y, t))dy \\ & + \lambda \int_0^t F(t, \tau)\mathcal{N}_3(\phi(x, \tau))d\tau, \end{aligned} \quad (2.5)$$

And the two well known types

1. Integro-differential equations

2. Fractional Partial Integro-differential equations

There are various forms of nonlinear integral equations [58] as follows:

Urysohn-second kind	$\phi(t) = f(t) + \int_a^b K(t, s, \phi(s))ds$
Hammerstein	$\phi(t) = f(t) + \int_a^b K(t, s)f(s, \phi(s))ds$
Urysohn-Volterra	$\phi(t) = f(t) + \int_a^t K(t, s, \phi(s))ds$
Hammerstein-Volterra-second kind	$\phi(t) = f(t) + \int_a^t K(t, s)\mathcal{N}(s, \phi(s))ds$
Hammerstein-Volterra-first kind	$\int_a^t K(t, s)\mathcal{N}(s, \phi(s))ds = f(t)$
Chandrasekhar-H equation	$1 + \lambda \int_a^b [t\phi(t)\phi(s)/(t+s)]ds = f(t)$
Cauchy singular	$a(t)\phi(t) + b(t) \int_{\Gamma} [\phi(s)/(s-t)]ds$ $+ \int_{\Gamma} K(t, s, \phi(s))ds = f(t)$ $\Gamma = \text{open or closed arc in } R^2$

2.3 The relation between initial and boundary value problems with IEs

It is important to point out how to convert the initial (boundary) value problems to Volterra (Fredholm) integral equations and vice versa [68, 8]. The following two theorems are needed while converting between these types of equations.

Theorem 2.1. *Leibnitz rule* [8]

Assume that $f(s, t)$ and its derivative $\frac{\partial f}{\partial t}$ are continuous in a domain on the plane st where $A(s) \leq s \leq B(s)$ and $t_0 \leq t \leq t_1$. If

$$\phi(s) = \int_{A(s)}^{B(s)} f(s, t) dt, \quad (2.6)$$

then the differentiation of Eq.(2.6) is given by

$$\frac{d\phi(s)}{ds} = \int_{A(s)}^{B(s)} \frac{\partial f(s, t)}{\partial s} dt + f(s, B(s)) \frac{dB}{ds} - f(s, A(s)) \frac{dA}{ds}. \quad (2.7)$$

Theorem 2.2. *The general formula that converts multiple integrals to a single integral is given in [8] by*

$$\int_a^x \int_a^{x_1} \cdots \int_a^{x_{n-1}} u(x_n) dx_n dx_{n-1} \cdots dx_1 = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} u(t) dt \quad (2.8)$$

where n is positive integer.

Next, we proceed how to convert IVPs to VIEs and vice versa

Converting IVPs to VIEs [68, 8]

Consider the following IVP

$$y''(x) + p(x)y'(x) + g(x)y(x) = h(x), \quad (2.9)$$

with the initial conditions $y(0) = \alpha$ and $y'(0) = \beta$ where α and β are constants. Suppose

$$y''(x) = \phi(x), \quad (2.10)$$

integrating Eq.(2.10) on $[0, x]$ and using the initial conditions, we get

$$y'(x) = \beta + \int_0^x \phi(t)dt. \quad (2.11)$$

Similarly, integrating Eq.(2.11) on $[0, x]$ with the help of the initial conditions and Eq.(2.8), we obtain

$$y(x) = \alpha + \beta x + \int_0^x (x-t)\phi(t)dt. \quad (2.12)$$

Substituting from Eqs.(2.10), (2.11) and (2.12) for Eq.(2.9), we get

$$\phi(x) = h(x) - \beta p(x) - \alpha g(x) - \beta x g(x) - \int_0^x [p(x) + g(x)(x-t)]\phi(t)dt. \quad (2.13)$$

Let $f(x) = h(x) - \beta p(x) - (\alpha + x\beta)g(x)$ and $K(x, t) = p(x) + g(x)(x-t)$, then Eq.(2.13) can be written in the form

$$\phi(x) = f(x) - \int_0^x K(x, t)\phi(t)dt \quad (2.14)$$

which represents VIE of the second kind.

Converting VIEs to IVPs [68, 8]

Consider the following VIE of the second kind

$$\phi(x) = f(x) + \int_0^x K(x, t)\phi(t)dt \quad (2.15)$$

where

$$f(x) = h(x) - \beta p(x) - (\alpha + x\beta)g(x) \text{ and } K(x, t) = p(x) + g(x)(x - t).$$

Let $p(x) = \lambda_1$ and $g(x) = \lambda_2$ where λ_1 and λ_2 are constants, Eq.(2.15) becomes

$$\phi(x) = f(x) + \lambda_1 \int_0^x \phi(t)dt + \lambda_2 \int_0^x (x - t)\phi(t)dt. \quad (2.16)$$

Differentiating Eq.(2.16) with respect to x and using Eq.(2.7), we get

$$\phi'(x) = f'(x) + \lambda_1 \phi(x) + \lambda_2 \int_0^x \phi(t)dt. \quad (2.17)$$

Similarity, differentiating Eq.(2.17) with respect to x and using Leibnitz rule, we obtain

$$\phi''(x) = f''(x) + \lambda_1 \phi'(x) + \lambda_2 \phi(x). \quad (2.18)$$

Eq.(2.18) represents ODE of the second order.

Putting $x = 0$ in Eq.(2.16) and Eq.(2.17), we get the initial conditions

$$\phi(0) = f(0), \quad \phi'(0) = f'(0) + \lambda_1 f(0). \quad (2.19)$$

Therefore, Eq.(2.18) with the conditions (2.19) represents an IVP converted from the VIE (2.15).

Now, we present examples of the converting process from BVPs to FIEs and vice versa.

Converting BVPs to FIEs [68, 8]

Consider the following BVP

$$y''(x) + g_1(x)y'(x) + g_2(x)y(x) = H(f(x), y(x)), \quad (2.20)$$

with the boundary conditions $y(0) = \alpha$, $y(1) = \beta$. Let

$$y''(x) = \phi(x). \quad (2.21)$$

Integrating Eq.(2.21) two times on $[0, x]$ and using the boundary conditions, we obtain

$$\int_0^x y''(t)dt = \int_0^x \phi(t)dt \quad (2.22)$$

Eq.(2.22) can be expressed as

$$y'(x) = y'(0) + \int_0^x \phi(t)dt. \quad (2.23)$$

Integrating Eq.(2.23) on $[0, x]$, using Eq.(2.8) and boundary conditions, yield

$$y(x) = \alpha + xy'(0) + \int_0^x (x-t)\phi(t)dt. \quad (2.24)$$

Since $y(1) = \beta$, we get

$$y'(0) = \beta - \alpha - \int_0^1 (1-t)\phi(t)dt. \quad (2.25)$$

So, Eq.(2.23) becomes

$$y'(x) = \beta - \alpha - \int_0^1 (1-t)\phi(t)dt + \int_0^x \phi(t)dt. \quad (2.26)$$

Substituting from Eq.(2.25) and Eq.(2.24) yields

$$y(x) = \alpha + x(\beta - \alpha) - \int_0^1 x(1-t)\phi(t)dt + \int_0^x (x-t)\phi(t)dt. \quad (2.27)$$

Substituting from Eqs.(2.21), (2.26) and (2.27), thus Eq.(2.20) becomes

$$\begin{aligned} \phi(x) = & \tilde{H}(x) - g_1(x) \left[\beta - \alpha - \int_0^1 (1-t)\phi(t)dt + \int_0^x \phi(t)dt \right] \\ & - g_2(x) \left[\alpha + x(\beta - \alpha) - \int_0^1 x(1-t)\phi(t)dt + \int_0^x (x-t)\phi(t)dt \right], \end{aligned} \quad (2.28)$$

$$\text{where } \tilde{H}(x) = H \left(f(x), \alpha + x(\beta - \alpha) - \int_0^1 x(1-t)\phi(t)dt + \int_0^x (x-t)\phi(t)dt \right).$$

Eq.(2.28) can be simplified as

$$\begin{aligned} \phi(x) = & \tilde{H}(x) + g_1(x)(\alpha - \beta) - g_2(x)(\alpha + x(\beta - \alpha)) \\ & + \int_0^1 (g_1(x) + xg_2(x))(1-t)\phi(t)dt - \int_0^x (g_1(x) + (x-t)g_2(x))\phi(t)dt. \end{aligned} \quad (2.29)$$

Partitioning of $[0, 1]$ to $[0, x]$ and $[x, 1]$, we find that Dividing integration in Eq.(2.29) from 0 to 1, we get

$$\begin{aligned} \phi(x) = & \tilde{H}(x) + g_1(x)(\alpha - \beta) - g_2(x)(\alpha + x(\beta - \alpha)) \\ & + \int_0^x (g_1(x) + g_2(x))(1-t)\phi(t)dt \\ & + \int_x^1 (g_1(x) + g_2(x))(1-t)\phi(t)dt \\ & - \int_0^x (g_1(x) + (x-t)g_2(x))\phi(t)dt. \end{aligned} \quad (2.30)$$

Eq.(2.30) can be written in the form

$$\begin{aligned}\phi(x) = & \tilde{H}(x) + g_1(x)(\alpha - \beta) - g_2(x)(\alpha + x(\beta - \alpha)) \\ & + \int_0^x t(g_2(x)(1 - x) - g_1(x))\phi(t)dt \\ & + \int_x^1 (g_1(x) + xg_2(x))(1 - t)\phi(t)dt.\end{aligned}\tag{2.31}$$

Let $\tilde{f}(x) = \tilde{H}(x) + g_1(x)(\alpha - \beta) - g_2(x)(\alpha + x(\beta - \alpha))$ and

$$K(x, t) = \begin{cases} t(g_2(x)(1 - x) - g_1(x)), & \text{for } 0 \leq t \leq x \\ (g_1(x) + xg_2(x))(1 - t), & \text{for } x \leq t \leq 1 \end{cases}.$$

Eq.(2.31) can be written in the form

$$\phi(x) = \tilde{f}(x) + \int_0^1 K(x, t)\phi(t)dt,\tag{2.32}$$

which represents FIE of the second kind.

Converting FIEs to BVPs [68, 8]

Consider the following FIE of the second kind

$$\phi(x) = \tilde{f}(x) + \int_0^1 K(x, t)\phi(t)dt,\tag{2.33}$$

where $\tilde{f}(x) = \tilde{H}(x) + g_1(x)(\alpha - \beta) - g_2(x)(\alpha + x(\beta - \alpha))$ and

$$K(x, t) = \begin{cases} t(g_2(x)(1 - x) - g_1(x)), & 0 \leq t \leq x \\ (g_1(x) + xg_2(x))(1 - t), & x \leq t \leq 1 \end{cases}.$$

Partitioning the integral in Eq.(2.33) yield

$$\begin{aligned}\phi(x) = & \tilde{f}(x) + \int_0^x t(g_2(x)(1-x) - g_1(x))\phi(t)dt \\ & + \int_x^1 (g_1(x) + xg_2(x))(1-t)\phi(t)dt.\end{aligned}\tag{2.34}$$

From Eq.(2.7), we get

$$\begin{aligned}\frac{d}{dx} \int_0^x t(g_2(x)(1-x) - g_1(x))\phi(t)dt = & x((1-x)g_2(x) - g_1(x))\phi(x) \\ & + \int_0^x t((1-x)g_2'(x) - g_2(x) - g_1'(x))\phi(t)dt\end{aligned}\tag{2.35}$$

and

$$\begin{aligned}\frac{d}{dx} \int_x^1 (g_1(x) + xg_2(x))(1-t)\phi(t)dt = & (g_1(x) + xg_2(x))(x-1)\phi(x) \\ & + \int_x^1 (g_1'(x) + g_2(x) + xg_2'(x))(1-t)\phi(t)dt.\end{aligned}\tag{2.36}$$

Differentiating Eq.(2.34) with respect to x , using Eqs.(2.35) and (2.36), we obtain

$$\begin{aligned}\phi'(x) = & \tilde{f}'(x) - g_1(x)\phi(x) \\ & + \int_0^x t((1-x)g_2'(x) - g_2(x) - g_1'(x))\phi(t)dt \\ & + \int_x^1 (g_1'(x) + g_2(x) + xg_2'(x))(1-t)\phi(t)dt.\end{aligned}\tag{2.37}$$

Using Eq.(2.7), we have

$$\begin{aligned}
\frac{d}{dx} \int_0^x t((1-x)g_2'(x) - g_2(x) - g_1'(x))\phi(t)dt \\
= x((1-x)g_2'(x) - g_2(x) - g_1'(x))\phi(x) \\
+ \int_0^x t((1-x)g_2''(x) - 2g_2'(x) - g_1''(x))\phi(t)dt
\end{aligned} \tag{2.38}$$

and

$$\begin{aligned}
\frac{d}{dx} \int_x^1 (g_1'(x) + g_2(x) + xg_2'(x))(1-t)\phi(t)dt \\
= -(g_1'(x) + g_2(x) + xg_2'(x))(1-x)\phi(x) \\
+ \int_x^1 (g_1''(x) + 2g_2'(x) + xg_2''(x))(1-t)\phi(t)dt.
\end{aligned} \tag{2.39}$$

Differentiating both sides of Eq.(2.37) with respect to x with the help of Eqs.(2.38) and (2.39), we obtain

$$\phi''(x) = f''(x) - \lambda x\phi(x) + \lambda(1-x)\phi(x), \tag{2.40}$$

Differentiating Eq.(2.34) two times with respect to x with the help of (2.7), we obtain and (2.8), we obtain

$$\begin{aligned}
\phi''(x) = \tilde{f}''(x) - g_1(x)\phi'(x) - (2g_1'(x) + g_2(x))\phi(x) \\
+ \int_0^x t((1-x)g_2''(x) - 2g_2'(x) - g_1''(x))\phi(t)dt \\
+ \int_x^1 (g_1''(x) + 2g_2'(x) + xg_2''(x))(1-t)\phi(t)dt.
\end{aligned} \tag{2.41}$$

For simplicity, let $g_1(x) = 0$ and $g_2(x) = \lambda$. So, Eq.(2.41) becomes

$$\phi''(x) + \lambda\phi(x) = \tilde{f}''(x). \quad (2.42)$$

To determine the boundary conditions, substitute by $x = 0$ and $x = 1$ in Eq.(2.34), we find

$$\phi(0) = \tilde{f}(0) \text{ and } \phi(1) = \tilde{f}(1). \quad (2.43)$$

Therefore, Eq.(2.42) with conditions (2.43) represents BVP which converted from FIE (2.33).

2.4 Existence and uniqueness theorems

In this section, we present some properties of a class of contraction operators in addition to number of basic existence and uniqueness theorems for integral equations.

Definition 2.1. Contraction operator [16]

Let H be a Hilbert space and T a bounded operator on H . T is not necessarily a linear operator. T is said to be a contraction operator if there exists a positive constant $0 \leq \alpha < 1$ such that for all f_1, f_2 in H .

$$\|Tf_1 - Tf_2\| \leq \alpha \|f_1 - f_2\|$$

Definition 2.2. Fixed point [16]

Let X be a non-empty set and $T : X \rightarrow X$ a mapping, a fixed point of the mapping T is a point $x \in X$ such that $Tx = x$. In other words, a fixed point of T is a solution for the IE $(Tx = x) \forall x \in X$.

Theorem 2.3. Banach fixed point theorem [16].

Let H be a Hilbert space and T be a contraction operator on H . The equation

$$Tf = f \quad (2.44)$$

has a unique solution f in H . Such a solution is said to be a fixed point of T .

Proof. Assume there are two fixed points f and g therefore

$$Tf = f,$$

$$Tg = g.$$

Subsequently

$$\|f - g\| = \|Tf - Tg\| \leq \alpha \|f - g\|$$

and

$$(1 - \alpha)\|f - g\| \leq 0.$$

Since $\|f - g\| \geq 0$ and $1 - \alpha < 0$, we get

$$\|f - g\| = 0,$$

so $f = g$. It follows that the solution of Eq.(2.44) is unique.

To show that $\{f_n\}$ has a solution, we shall set up an iteration procedure. Select any f_0 and then construct a sequence $\{f_n\}$ defined by

$$f_{n+1} = Tf_n, \quad n = 0, 1, 2, \dots$$

Firstly, we shall show that $\{f_n\}$ is a Cauchy sequence, and then that its limit is indeed a solution of Eq.(2.44). That it has a limit will follow from the fact that a Cauchy sequence must have a unique limit in a Hilbert space. The limit will be independent of the initial choice f_0 , since it will be a solution of Eq.(2.44), which must be unique. Now, we note

$$\|f_{n+1} - f_n\| = \|Tf_n - Tf_{n-1}\| \leq \alpha \|f_n - f_{n-1}\|. \quad (2.45)$$

From Eq.(2.45), we have

$$\|f_{n+1} - f_n\| \leq \alpha \|f_n - f_{n-1}\| \leq \alpha^2 \|f_{n-1} - f_{n-2}\| \leq \dots \leq \alpha^n \|f_1 - f_0\|.$$

For $n > m$, we get

$$\begin{aligned}
 \|f_n - f_m\| &= \|(f_n - f_{n-1}) + (f_{n-1} - f_{n-2}) + \cdots + (f_{m+1} - f_m)\| \\
 &\leq \|f_n - f_{n-1}\| + \|f_{n-1} - f_{n-2}\| + \cdots + \|f_{m+1} - f_m\| \\
 &\leq (\alpha^{n-1} + \alpha^{n-2} + \cdots + \alpha^m) \|f_1 - f_0\| \\
 &\leq (\alpha^m + \alpha^{m+2} + \cdots) \|f_1 - f_0\| = \frac{\alpha^m}{1 - \alpha} \|f_1 - f_0\|
 \end{aligned}$$

so that

$$\lim_{n,m \rightarrow \infty} \|f_n - f_m\| = 0$$

It follows that $\{f_n\}$ is Cauchy sequence and its limit is denoted by f . $\square$

Theorem 2.4. [27]

Let T be an operator on H , such that the n^{th} power of T , namely T^n is a contraction operator, then the equation

$$Tf = f$$

has a unique solution f in H .

Proof. Using Theorem (2.4), we have

$$T^n f = f \tag{2.46}$$

has a unique solution. Actually, we can obtain the solution by finding

$$\lim_{k \rightarrow \infty} T^{kn} f_0 = f,$$

for an arbitrary initial function f_0 . In particular, let $f_0 = Tf$, we get

$$\lim_{k \rightarrow \infty} T^{kn} Tf = f.$$

But since $T^n f = f$, we also have $T^{kn} f = f$ so that $\lim_{k \rightarrow \infty} T^{kn} Tf = \lim_{k \rightarrow \infty} T T^{kn} f = \lim_{k \rightarrow \infty} Tf = Tf$.

To show that this solution is unique, we note that if

$$Tf = f, \quad Tg = g,$$

we have

$$T^n f = f, \quad T^n g = g$$

and since T^n is a contraction operator with a unique fixed point $f = g$. $\square$

An existence theorem for nonlinear Fredholm integral equations[27]

Theorem 2.5. [27]

Consider the following nonlinear Fredholm integral equation

$$\phi(x) - \lambda \int_a^b K(x, y, \phi(y)) dy = f(x), \quad (2.47)$$

satisfies

$$\left\| \int_a^b K(x, y, \phi(y)) dy \right\| \leq M \|\phi(y)\|$$

and

$$|K(x, y, \phi_1(y)) - K(x, y, \phi_2(y))| \leq N(x, y) |\phi_1(y) - \phi_2(y)|,$$

where

$$\int_a^b \int_a^b |N(x, y)|^2 dx dy = P^2 < \infty.$$

If the function $f(x) \in L_2[a, b]$ and $|\lambda|P < 1$, then Eq (2.47) has a unique solution.

An existence theorem for nonlinear Volterra integral equations[?]

Theorem 2.6. [8]

Suppose the following nonlinear Volterra integral equation

$$\phi(x) + \int_a^x K(x, y, \phi(y)) dy = f(x), \quad (2.48)$$

satisfies the following assumptions:

- i. the function $f(x)$ is bounded and satisfies Lipschitz condition, i.e. $|f(x)| < f$, in the interval $a \leq x \leq b$ and satisfies Lipschitz condition:

$$|f(x_1) - f(x_2)| < l |x_1 - x_2| \quad x_1, x_2 \in (a, b). \quad (2.49)$$

- ii. the function $K(x, y, \phi(y))$ is integrable and bounded,

$$|K(x, y, \phi(y))| < K, \quad a \leq x, y \leq b.$$

and

- iii. the function $K(x, y, \phi(x))$ satisfies Lipschitz condition, such that:

$$|K(x, y, \phi_1(y)) - K(x, y, \phi_2(y))| < M |\phi_1(y) - \phi_2(y)| \quad (2.50)$$

then Eq.(2.48) has a unique solution.

2.5 Kernels of IEs

We have two types of kernels; continuous and discontinuous kernels.

1. Degenerate kernel

The kernel $k(x, t)$ of IE is called degenerate kernel if it is a sum of finite number of product functions of x only and t only

$$k(x, t) = \sum_{i=1}^n a_i(x) b_i(t)$$

where $a_i(x)$ and $b_i(t)$ are sets of linearly independent functions.

2. Resolvent kernel

The resolvent kernel $R(x, t; \lambda)$ are defined as

$$R(x, t; \lambda) = \frac{D(x, t; \lambda)}{\Delta(\lambda)}$$

where $D(x, t; \lambda) = \sum_{n=1}^{\infty} \lambda^{n-1} k_n(x, t)$, $\Delta(\lambda) = 1 + \sum_{n=1}^{\infty} c_n \lambda^n$,
 $k_n(x, t) = \int_a^b k_{n-1}(x, s)k(s, t)ds$ and $k_1(x, t) = k(x, t)$.

3. Iterated kernel

The iterated kernels $k_n(x, t)$, $n = 1, 2, 3, \dots$ are defined as

$$k_n(x, t) = \int_a^b k(x, z)k_{n-1}(z, t)dz, \quad k_0(x, t) = k(x, t).$$

4. Symmetric kernel

A kernel $k(x, t)$ is called symmetric (Hermitian) if

$$k(s, t) = k^*(t, s)$$

where $k^*(t, s)$ denotes the complex conjugate.

5. Orthogonal kernels

Assume that two kernels $k(x, t)$ and $L(x, t)$ are given, these kernels are said to be orthogonal if the following two conditions

$$\int_a^b k(x, z)L(z, t)dz = 0 \quad \text{and} \quad \int_a^b L(x, z)k(z, t)dz = 0,$$

for all values of x and t are satisfied.

2.5.1 (discontinuous) kernels

A singular kernel is a kernel that has an infinite discontinuity in the interior of the interval of integration or at a boundary point of it. The singular kernel may take one of the following forms:

1. Cauchy kernel

If the kernel $k(x, t)$ of IE takes the form

$$k(x, t) = \frac{A(x, t)}{x - t}$$

where $A(x, t) \neq 0$ is a differentiable function of x and t .

2. Abel's kernel

If the kernel $k(x, t)$ of IE takes the form

$$k(x, t) = \frac{A(x, t)}{(x - t)^n}, \quad 0 < n < 1.$$

The IE is called a singular IE with Abel's kernel.

3. Carleman kernel

If the kernel $k(x, t)$ of IE takes the form

$$k(x, t) = \frac{A(x, t)}{|x - t|^n}, \quad 0 < n < 1.$$

The IE is called a singular IE with Carleman kernel.

4. Logarithmic kernel

If the kernel $k(x, t)$ of IE takes the form

$$k(x, t) = A(x, t) \ln |x - t|.$$

The IE is called a singular IE with logarithmic kernel.

2.6 Analytical methods for solving IEs with continuous kernel

Now, we are going to discuss some analytical methods [68, ?, ?] which depend on the form of kernel of IEs; we can use the DKM if the kernel is continuous and can be written as multiplication of two functions. Otherwise, we can use the SAM. When the kernel is in the form $k(x - t)$, we can use Laplace transformation and convolution theorem to solve IEs.

2.6.1 Degenerate kernel method

Consider the following FIE of the second kind

$$\phi(x) = f(x) + \lambda \int_a^b k(x, t) \phi(t) dt. \quad (2.51)$$

Assume that the kernel $k(x, t) \in L_2([a, b] \times [a, b])$ and is bounded such that

$$k(x, t) = \sum_{i=1}^n a_i(x)b_i(t). \quad (2.52)$$

Substituting from Eq.(2.52) in Eq.(2.51) and interchanging the order of integration and summation, we get

$$\phi(x) = f(x) + \lambda \sum_{i=1}^n a_i(x) \int_a^b b_i(t)\phi(t)dt. \quad (2.53)$$

$$\text{Let } c_i = \int_a^b b_i(t)\phi(t)dt, \quad 1 \leq i \leq n \quad (2.54)$$

where c_i are arbitrary constants. Substituting from Eq.(2.54) in Eq.(2.53) yields

$$\phi(x) = f(x) + \lambda \sum_{i=1}^n a_i(x)c_i. \quad (2.55)$$

Substituting from (2.55) in (2.54), we get

$$c_i = \int_a^b b_i(t) \left[f(t) + \lambda \sum_{j=1}^n a_j(t)c_j \right] dt. \quad (2.56)$$

$$c_i - \lambda \sum_{j=1}^n c_j \int_a^b a_j(t)b_i(t)dt = \int_a^b b_i(t)f(t)dt, \quad (2.57)$$

$$\text{Let } A_{ij} = \int_a^b a_j(t)b_i(t)dt \text{ and } B_i = \int_a^b b_i(t)f(t)dt, \quad (2.58)$$

then Eq.(2.57) can be represented in the form

$$c_i - \lambda \sum_{j=1}^n c_j A_{ij} = B_i, \quad 1 \leq i \leq n. \quad (2.59)$$

The system (2.59) represents an algebraic system of n unknown constants $c_1, c_2, \dots, c_n$. Finding the solution of FIE (2.51) with DKM is equivalent to find the solution of the system (2.59)

$$(I - \lambda A)C = B \quad (2.60)$$

where

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & & & \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{pmatrix}, \quad C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \text{ and } B = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{bmatrix}. \quad (2.61)$$

The system (2.60) has a unique solution if $\text{Det}(I - \lambda A) \neq 0$ and either no solution or infinite solutions if $\text{Det}(I - \lambda A) = 0$. After we calculate the unknown constants $c_1, c_2, \dots, c_n$, we substitute in (2.55) which represents the solution of FIE (2.51) of the second kind.

Example 2.7. Solve the following FIE of the second kind using DKM

$$\phi(x) = 4x + \int_0^1 (x \ln t - t \ln x) \phi(t) dt. \quad (2.62)$$

Solution From Eq.(2.52), we get $a_1(x) = x$, $a_2(x) = \ln x$, $b_1(t) = \ln t$ and $b_2(t) = -t$. Eq.(2.58) yields

$$A_{11} = -\frac{1}{4}, A_{12} = 2, A_{21} = -\frac{1}{3}, A_{22} = \frac{1}{4}, B_1 = -1 \text{ and } B_2 = -\frac{4}{3}. \quad (2.63)$$

$$\text{Since } \text{Det}(I - \lambda A) = \frac{7}{48} \neq 0, \quad (2.64)$$

the system (2.60) has a unique solution. Substituting from Eq.(2.63) in Eq.(2.60) yields

$$\begin{pmatrix} -\frac{1}{4} & 2 \\ -\frac{1}{3} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -\frac{4}{3} \end{pmatrix}. \quad (2.65)$$

The solution of the system (2.65) is

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -\frac{164}{77} \\ -\frac{64}{77} \end{pmatrix}. \quad (2.66)$$

Substituting from Eq.(2.66) in Eq.(2.55) yields

$$\begin{aligned} \phi(x) &= 4x + \sum_{i=1}^2 a_i(x)c_i \\ &= 4x + x\left(-\frac{164}{77}\right) + \ln x\left(-\frac{64}{77}\right) = \frac{144}{77}x - \frac{64}{77}\ln x. \end{aligned} \quad (2.67)$$

Theorem 2.8. [?]

If $f(x)$ in the recurrence relation

$$\phi_n(x) = f(x) + \lambda \int_0^x K(x, t)\phi_{n-1}(t)dt, \quad n \geq 1 \quad (2.68)$$

in the interval $0 \leq x \leq a$ and the kernel $K(x, t)$ in the triangle $0 \leq x \leq a$ and $0 \leq t \leq a$ are continuous, then the sequence of successive approximations $\phi_n(x), n \geq 0$ converges to the solution $\phi(x)$ of IE under discussion.

2.6.2 Successive approximation method[?]

We are going to introduce the SAM for solving FIEs of the second kind. The SAM is called Picard's iteration method. Using this method depends on finding the successive approximations to the solution starting with an initial guess which is called the zeroth approximation. Moreover, the zeroth approximation is any selective real-valued function that will be used in a recurrence relation to determine other approximations.

Assume that $f(x)$ and the kernel $k(x, t)$ are continuous in the interval $[a, b]$ in Eq.(2.51) and choose continuous function which is called the zeroth approxima-

tion $\phi_0(x) = f(x)$. The SAM introduces the recurrence relation

$$\phi_{n+1}(x) = f(x) + \lambda \int_a^b k(x, t) \phi_n(t) dt. \quad (2.69)$$

Several successive approximations will be determined as

$$\phi_1(x) = f(x) + \lambda \int_a^b k(x, t) \phi_0(t) dt. \quad (2.70)$$

Using Eq.(2.70), Eq.(2.69) yields the second-order approximation

$$\phi_2(x) = f(x) + \lambda \int_a^b k(x, t) \phi_1(t) dt. \quad (2.71)$$

Continuing the process, we obtain

$$\phi_n(x) = f(x) + \lambda \int_a^b k(x, t) \phi_{n-1}(t) dt. \quad (2.72)$$

Using theorem (2.8) and definition of convergence, the solution of the IE is determined by using a limit of iterations as

$$\phi(x) = \lim_{n \rightarrow \infty} \phi_n(x). \quad (2.73)$$

Eq.(2.73) represents the solution of FIE (2.51), using SAM.

Example 2.9. Using SAM to solve the following FIE of the second kind

$$\phi(x) = e^x + \frac{1}{e} \int_0^1 \phi(t) dt. \quad (2.74)$$

Solution Let $\phi_0(x) = f(x) = e^x$ and substituting in Eq.(2.72), we obtain

$$\begin{aligned}\phi_1(x) &= e^x + 1 - \frac{1}{e}, & \phi_2(x) &= e^x + 1 - \frac{1}{e^2}, \\ \phi_3(x) &= e^x + 1 - \frac{1}{e^3} \text{ and } \phi_n(x) &= e^x + 1 - \frac{1}{e^n}.\end{aligned}\tag{2.75}$$

Therefore, the solution of Eq.(2.74) is a limit of iterations s.t.

$$\phi(x) = \lim_{n \rightarrow \infty} \phi_n(x) = \lim_{n \rightarrow \infty} \left\{ e^x + 1 - \frac{1}{e^n} \right\} = e^x + 1.\tag{2.76}$$

Theorem 2.10. [?]

Let λ be a complex parameter and $f(x)$ and $k(x, t)$ a complex-valued continuous functions defined on the interval $[a, b]$ with

$$\|k\|_2 = \left(\int_a^b \int_a^b |k(x, t)|^2 dx dt \right)^{1/2}.\tag{2.77}$$

If $|\lambda| \|k\|_2 < 1$, then the unique solution to the FIE of the second kind

$$\phi(x) = f(x) + \lambda \int_a^b k(x, t) \phi(t) dt\tag{2.78}$$

is given by

$$\phi(x) = f(x) + \lambda \int_a^b R(x, t; \lambda) f(t) dt\tag{2.79}$$

where $R(x, t; \lambda)$ is the resolvent kernel

$$R(x, t; \lambda) = \sum_{n=1}^{\infty} \lambda^{n-1} k_n(x, t)\tag{2.80}$$

where $k_n(x, t) = \int_a^b k_{n-1}(x, s) k(s, t) ds$.

2.6.3 Resolvent kernel method

Consider the VIE of the second kind

$$\phi(x) = f(x) + \lambda \int_0^x k(x, t) \phi(t) dt \quad (2.81)$$

where $k(x, t)$ is continuous for $0 \leq x \leq a, 0 \leq t \leq x$ and $f(x)$ is continuous for $0 \leq x \leq a$. Let $\phi_0(x) = f(x)$ and

$$\phi_n(x) = f(x) + \lambda \int_0^x k(x, t) \phi_{n-1}(t) dt, \quad (2.82)$$

we get

$$\phi_{n-1}(x) = f(x) + \lambda \int_0^x k(x, t) \phi_{n-2}(t) dt. \quad (2.83)$$

Subtracting Eq.(2.83) from Eq.(2.82) yields

$$\phi_n(x) - \phi_{n-1}(x) = \lambda \int_0^x k(x, t) [\phi_{n-1}(t) - \phi_{n-2}(t)] dt. \quad (2.84)$$

Assume that

$$\phi_n(x) - \phi_{n-1}(x) = \lambda^n \psi_n(t) \text{ and } \psi_0(x) = f(x), \quad (2.85)$$

substituting from Eq.(2.85) in Eq.(2.84) yields

$$\psi_n(x) = \int_0^x k(x, t) \psi_{n-1}(t) dt. \quad (2.86)$$

From Eq.(2.86), we have

$$\psi_1(x) = \int_0^x k(x, t) f(t) dt \quad (2.87)$$

$$\text{and } \psi_2(x) = \int_0^x k(x, t)\psi_1(t)dt = \int_0^x k(x, t) \left[\int_0^t k(x, \tau)f(\tau)d\tau \right] dt. \quad (2.88)$$

Since $k(x, t)$ and $f(x)$ are continuous, the order of integration in Eq.(2.88) can be interchanged s.t.

$$\begin{aligned} \psi_2(x) &= \int_0^x \int_\tau^x k(x, t)k(t, \tau)f(\tau)dtd\tau \\ &= \int_0^x k_2(x, \tau)f(\tau)d\tau \end{aligned} \quad (2.89)$$

where

$$k_2(x, \tau) = \int_\tau^x k(x, t)k(t, \tau)dt. \quad (2.90)$$

Similarly, we have

$$\psi_n(x) = \int_0^x k_n(x, t)f(t)dt \quad (2.91)$$

where $k_n(x, t)$ are called iterative kernels and determine from the relation

$$k_n(x, t) = \int_t^x k(x, \tau)k_{n-1}(\tau, t)d\tau, \text{ and } k_1(x, t) = k(x, t). \quad (2.92)$$

Eq.(2.85) yields

$$\begin{aligned} \phi_1(x) &= \psi_0(x) + \lambda\psi_1(x), \\ \phi_2(x) &= \psi_0(x) + \lambda\psi_1(x) + \lambda^2\psi_2(x) \text{ and} \\ \phi_3(x) &= \psi_0(x) + \lambda\psi_1(x) + \lambda^2\psi_2(x) + \lambda^3\psi_3(x). \end{aligned} \quad (2.93)$$

From system (2.93), we get

$$\phi_n(x) = \sum_{i=0}^n \lambda^i \psi_i(x). \quad (2.94)$$

Substituting from Eq.(2.91) in Eq.(2.94), we obtain

$$\begin{aligned}
 \phi_n(x) &= \psi_0(x) + \sum_{i=1}^n \lambda^i \psi_i(x) \\
 &= f(x) + \lambda \int_0^x \sum_{i=1}^n \lambda^{i-1} k_i(x, t) f(t) dt \\
 &= f(x) + \lambda \int_0^x R(x, t, \lambda) f(t) dt
 \end{aligned} \tag{2.95}$$

where

$$R(x, t, \lambda) = \sum_{n=1}^{\infty} \lambda^{n-1} k_n(x, t). \tag{2.96}$$

Eq.(2.95) represents the solution of VIE (2.81) where the resolvent kernel is evaluated from Eq.(2.96).

Example 2.11. Using the resolvent kernel method to solve the following VIE of the second kind

$$\phi(x) = e^x + \int_0^x e^{x-t} \phi(t) dt. \tag{2.97}$$

Solution Assume that $k(x, t) = k_1(x, t) = e^{x-t}$, Eq.(2.92) yields

$$\begin{aligned}
 k_2(x, t) &= \int_t^x k(x, \tau) k_1(\tau, t) d\tau \\
 &= \int_t^x e^{x-\tau} d\tau = (x-t)e^{x-t}.
 \end{aligned} \tag{2.98}$$

Using the values of $k_1(x, t)$ and $k_2(x, t)$, we obtain

$$k_3(x, t) = \int_t^x k(x, \tau) k_2(\tau, t) d\tau = \frac{(x-t)^2}{2} e^{x-t} \text{ and} \tag{2.99}$$

$$k_4(x, t) = \int_t^x k(x, \tau) k_3(\tau, t) d\tau = \frac{(x-t)^3}{6} e^{x-t}. \tag{2.100}$$

From Eq.(2.98), Eq.(2.99) and Eq.(2.100), we get

$$\begin{aligned} k_n(x, t) &= e^{x-t} + (x-t)e^{x-t} + \frac{(x-t)^2}{2}e^{x-t} + \frac{(x-t)^3}{6}e^{x-t} + \dots \\ &= \frac{(x-t)^{n-1}}{(n-1)!}e^{x-t}. \end{aligned} \quad (2.101)$$

Eq.(2.96) yields that the resolvent kernel is

$$\begin{aligned} R(x, t, \lambda) &= k_1(x, t) + k_2(x, t) + k_3(x, t) + \dots \\ &= \sum_{n=1}^{\infty} \frac{(x-t)^{n-1}}{(n-1)!}e^{x-t} \\ &= e^{2(x-t)}. \end{aligned} \quad (2.102)$$

Substituting from Eq.(2.102) in Eq.(2.95), the solution of Eq.(2.97) is

$$\phi(x) = e^x + \int_0^x e^{2(x-t)}e^t dt = e^x + \int_0^x e^{2x-t} dt = e^{2x}. \quad (2.103)$$

Theorem 2.12. Fredholm alternative theorem[?]

If the homogeneous FIE of the second kind

$$\phi(x) = \lambda \int_a^b k(x, t)\phi(t)dt \quad (2.104)$$

has unique solution $\phi(x) = 0$, then the corresponding nonhomogeneous FIE

$$\phi(x) = f(x) + \lambda \int_a^b k(x, t)\phi(t)dt \quad (2.105)$$

has always a unique solution. The adjoint equation

$$\psi(x) = \bar{\lambda} \int_a^b \overline{k(y, x)}\psi(y)dy \quad (2.106)$$

has the same number of solutions of Eq.(2.104). If the number of solutions of

Eq.(2.104) and Eq.(2.106) is positive, then Eq.(2.105) will have a nonunique solution, iff

$$(f, \psi) = \int_a^b f(x) \overline{\psi(x)} dx = 0 \quad (2.107)$$

for all solutions $\psi(x)$ of Eq.(2.106)

Example 2.13. Consider the following FIE of the second kind

$$\phi(x) = f(x) + \lambda \int_0^1 [\pi x \sin \pi t + 2\pi x^2 \sin 2\pi t] \phi(t) dt. \quad (2.108)$$

Solution Let

$$\begin{aligned} c_1 &= \int_0^1 \sin \pi t \phi(t) dt, & B_1 &= \int_0^1 \sin \pi t f(t) dt \\ c_2 &= \int_0^1 \sin 2\pi t \phi(t) dt \text{ and } & B_2 &= \int_0^1 \sin 2\pi t f(t) dt, \end{aligned} \quad (2.109)$$

from the system (2.59), Eq.(2.108) can be reduced to

$$\begin{aligned} (1 - \lambda)c_1 - 2 \left(1 - \frac{4}{\pi^2}\right) \lambda c_2 &= B_1 \\ \frac{\lambda}{2} c_1 + (1 - \lambda)c_2 &= B_2 \end{aligned} \quad (2.110)$$

and from the systems (2.60) and (2.110), we get $\text{Det}(I - \lambda A) = \lambda^2 - \pi^2/4$.

So, we have:

(i) If $\lambda^2 \neq (\pi^2/4)$, we get that the system (2.110) has a unique solution s.t.

$$\begin{aligned} c_1 &= \frac{(1 + \lambda)B_1 + 2 \left[1 - (4/\pi^2)\right] \lambda B_2}{1 - (4\lambda^2/\pi^2)} \text{ and} \\ c_2 &= \frac{(-\lambda/2)B_1 + (1 - \lambda)B_2}{1 - (4\lambda^2/\pi^2)}, \end{aligned} \quad (2.111)$$

so the solution of the Eq.(2.108) is

$$\phi(x) = f(x) + \pi \lambda c_1 x + 2\pi \lambda c_2 x^2. \quad (2.112)$$

(ii) If $\lambda = (\pi/2)$, we consider the adjoint system

$$\left(1 - \frac{\pi}{2}\right) \zeta_1 + \frac{\pi}{4} \zeta_2 = 0, \quad -\pi \left(1 - \frac{4}{\pi^2}\right) \zeta_1 + \left(1 + \frac{\pi}{2}\right) \zeta_2 = 0 \quad (2.113)$$

with solutions $\zeta_1 = (\pi/4)\Lambda$ and $\zeta_2 = -[1 - (\pi/2)]\Lambda$ where Λ is arbitrary.

In order for a solution to exist, we require that

$$\frac{\pi}{4} B_1 - \left(1 - \frac{\pi}{2}\right) B_2 = 0.$$

If the above condition holds, we obtain

$$\phi(x) = f(x) + \frac{\pi}{2} \left\{ \frac{B_1}{1 - (\pi/2)} - 2 \left(1 + \frac{2}{\pi}\right) \Lambda x + \pi \Lambda x^2 \right\}. \quad (2.114)$$

Since Λ is arbitrary, the nonunique solution exists.

2.6.4 Laplace Transformation Method

We are going to introduce LTM for solving VIEs of the second kind which be a powerful technique that can be used for solving IEs. We can use this method when the kernel $k(x - t)$ of the IE is a function of the difference only, in this case the IE is called IE of convolution type.

Definition 2.3. Laplace transform

Let $f(x)$ be a function defined for $0 \leq x < \infty$, then the Laplace transform of $f(x)$ is denoted by $\mathcal{L}\{f(x); t\}$ or $F(t)$ and defined as

$$\mathcal{L}\{f(x); t\} = F(t) = \int_0^{\infty} e^{-tx} f(x) dx, \quad t > 0. \quad (2.115)$$

Laplace transform satisfies the linearity property.

If for $i \in \{1, 2, \dots, n\}$, c_i are constants and $f_i(x)$ are functions with Laplace transforms $F_i(t)$, then we have

$$\begin{aligned} \mathcal{L}\{c_1 f_1(x) + \dots + c_n f_n(x); p\} &= c_1 \mathcal{L}\{f_1(x); p\} + \dots + c_n \mathcal{L}\{f_n(x); p\} \\ &= c_1 F_1(p) + \dots + c_n F_n(p). \end{aligned} \quad (2.116)$$

Theorem 2.14. The convolution theorem

Let $f(x)$ and $g(x)$ be two functions with $\mathcal{L}^{-1}\{F(t); x\} = f(x)$ and $\mathcal{L}^{-1}\{G(t); x\} = g(x)$, then we get

$$\begin{aligned} \mathcal{L} \left\{ \int_0^x f(u)g(x-u)du \right\} &= \mathcal{L} \left\{ \int_0^x f(x-u)g(u)du \right\} \\ &= F(t)G(t). \end{aligned} \quad (2.117)$$

Applying the convolution definition and Laplace transform to Eq.(2.81) yield

$$\mathcal{L}\{\phi(x)\} = F(t) + \lambda K(t)\mathcal{L}\{\phi(x)\}. \quad (2.118)$$

Eq.(2.118) can be written in the form

$$\mathcal{L}\{\phi(x)\} = \frac{F(t)}{1 - \lambda K(t)}. \quad (2.119)$$

Taking the inverse Laplace transform to Eq.(2.119), we get

$$\phi(x) = \mathcal{L}^{-1}\left\{\frac{F(t)}{1 - \lambda K(t)}\right\}. \quad (2.120)$$

Eq.(2.120) represents the solution of VIE (2.81).

Example 2.15. Using LTM to solve the following VIE of the second kind

$$\phi(x) = x + \int_0^x \sin(x-t)\phi(t)dt. \quad (2.121)$$

Solution From Eq.(2.120), we obtain

$$\begin{aligned} \phi(x) &= \mathcal{L}^{-1}\left\{\frac{\mathcal{L}\{x\}}{1 - \mathcal{L}\{\sin(x)\}}\right\} \\ &= \mathcal{L}^{-1}\left\{\frac{\frac{1}{t^2}}{1 - \frac{1}{t^2+1}}\right\} \\ &= \mathcal{L}^{-1}\left\{\frac{1}{t^2} + \frac{1}{t^4}\right\} \\ &= x + \frac{x^3}{6}. \end{aligned} \quad (2.122)$$

2.7 Numerical methods

It is worth noting that IEs often do not have analytical solution, and must be solved numerically. So, many researchers have been done to get the Appr. solution of IEs, using different methods [26, ?, ?, ?]. In the next chapters, we concentrate our interest on illustrating the Appr. solutions.

Numerical analysis is the study of algorithms that use numerical approximation for the problems of mathematical analysis. Numerical analysis naturally finds applications in many fields such as engineering, physical sciences, life sciences, social sciences, medicine, and business. The growth in computing power has revolutionized the use of realistic mathematical models in science and engineering. For example, ordinary differential equations appear in celestial mechanics (predicting the motions of planets, stars, and galaxies); numerical linear algebra is important for data analysis; stochastic differential equations and Markov chains are essential in simulating living cells for medicine and biology. Numerical solution of partial differential equation is almost an important topic in recent years. partial differential equation arise in formulations of problems involving functions of several variables such as the propagation of sound or heat, electrostatics, electrodynamics, fluid flow, and elasticity, etc. Most Mathematical Models which contain differential equations, partial differential equations, integral equations, fractions equations, and mixed equations are solved numerically by different methods as finite difference method, Rang- kota method, variation iteration method, Adomain decomposition method [91, 17, 22, 23], Laplace Adomain decomposition method, homotopy analysis method [2, 15, 79, 82], homotopy perturbation method [25, 42, 62], Laplace homotopy perturbation method [50], and other methods. Therefore, we will review some numerical methods.

2.7.1 Quadrature methods for FIE and VIE

Using quadrature rule, we get

$$\int_a^b k(x, t)\phi(t)dt = \sum_{j=0}^n \omega_j k(x, t_j)\phi(t_j) \quad (2.123)$$

where ω_j are the weight functions.

Substituting from Eq.(2.123) in Eq.(??) yields

$$\mu\phi(x) = f(x) + \lambda \sum_{j=0}^n \omega_j k(x, x_j) \phi(x_j). \quad (2.124)$$

The solution for Eq.(2.124) may be obtained if we assign x_i to x which $i = 0, 1, 2, \dots, n$, $a \leq x_i \leq b$, so Eq.(2.124) reduced to a system of equations

$$\mu\phi(x_i) - \lambda \sum_{j=0}^n \omega_j k(x_i, x_j) \phi(x_j) = f(x_i). \quad (2.125)$$

For any $x_i \in [a, b]$, Eq.(2.125) can be represented by

$$(I - \frac{\lambda}{\mu} k D) \tilde{\phi}_i = \frac{1}{\mu} \tilde{f} \quad (2.126)$$

where $\tilde{\phi}_i = [\phi(x_i)]^T$, $\tilde{f} = [f(x_i)]^T$, $k = [k(x_i, x_j)]^T$ and $D = \text{diag}[\omega_0, \omega_1, \dots, \omega_n]$.

Trapezoidal rule for FIEs and VIEs

The TR is a numerical method used for solving FIE (??) of the second kind. We shall subdivide the interval of integration $[a, b]$ into n equal subintervals of width h_t where $h_t = \frac{b-a}{n}$. The general form of TR is

$$\int_a^b k(x, t) \phi(t) dt = \frac{h_t}{2} \left[k(x, t_0) \phi(t_0) + 2 \sum_{j=1}^{n-1} k(x, t_j) \phi(t_j) + k(x, t_n) \phi(t_n) \right]. \quad (2.127)$$

At $x = x_i$, $i = 0, 1, 2, \dots, n$, so substituting from Eq.(2.127) in Eq.(??) yields

$$\mu\phi(x_i) = f(x_i) + \frac{\lambda h_t}{2} \left[k(x_i, t_0) \phi(t_0) + 2 \sum_{j=1}^{n-1} k(x_i, t_j) \phi(t_j) + k(x_i, t_n) \phi(t_n) \right]. \quad (2.128)$$

Eq.(2.128) can be written in the form

$$\mu\phi_i = f_i + \frac{\lambda h_t}{2} \left[k_{i,0} \phi_0 + 2 \sum_{j=1}^{n-1} k_{i,j} \phi_j + k_{i,n} \phi_n \right] \quad (2.129)$$

where $\phi_i = \phi(x_i)$, $f_i = f(x_i)$ and $k_{i,n} = k(x_i, x_n)$.

Also, if the same method is used for solving VIE of the second kind, we get

$$\mu\phi_i = f_i + \frac{\lambda h_t}{2} \left[k_{i,0}\phi_0 + 2 \sum_{j=1}^{i-1} k_{i,j}\phi_j + k_{i,i}\phi_i \right]. \quad (2.130)$$

Simpson's rule for FIEs and VIEs

The Simpson's 1/3 rule is a numerical method used to solve FIE (??) of the second kind. We shall subdivide the interval of integration $[a, b]$ into positive n equal subintervals of width h_s where $h_s = \frac{b-a}{n}$ where n is even. The general form of Simpson's 1/3 rule is

$$\begin{aligned} \int_a^b k(x, t)\phi(t)dt &= \frac{h_s}{3} [k(x, t_0)\phi(t_0) + 4 \sum_{j=1}^{\frac{n}{2}} k(x, t_{2j-1})\phi(t_{2j-1}) \\ &\quad + 2 \sum_{j=1}^{\frac{n}{2}-1} k(x, t_{2j})\phi(t_{2j}) + k(x, t_n)\phi(t_n)]. \end{aligned} \quad (2.131)$$

We shall set $t_i = x_i$, $i = 0, 1, 2, \dots, n$, so Eq.(2.131) becomes

$$\begin{aligned} \int_a^b k(x, t)\phi(t)dt &= \frac{h_s}{3} [k(x, x_0)\phi(x_0) + 4 \sum_{j=1}^{\frac{n}{2}} k(x, x_{2j-1})\phi(x_{2j-1}) \\ &\quad + 2 \sum_{j=1}^{\frac{n}{2}-1} k(x, x_{2j})\phi(x_{2j}) + k(x, x_n)\phi(x_n)]. \end{aligned} \quad (2.132)$$

Substituting from Eq.(2.132) in Eq.(??) yields

$$\begin{aligned} \mu\phi(x) &= f(x) + \frac{\lambda h_s}{3} [k(x, x_0)\phi(x_0) + 4 \sum_{j=1}^{\frac{n}{2}} k(x, x_{2j-1})\phi(x_{2j-1}) \\ &\quad + 2 \sum_{j=1}^{\frac{n}{2}-1} k(x, x_{2j})\phi(x_{2j}) + k(x, x_n)\phi(x_n)]. \end{aligned} \quad (2.133)$$

At $x = x_i$, $i = 0, 1, 2, \dots, n$, Eq.(2.133) becomes

$$\begin{aligned} \mu\phi(x_i) &= f(x_i) + \frac{\lambda h_s}{3} [k(x_i, x_0)\phi(x_0) + 4 \sum_{j=1}^{\frac{n}{2}} k(x_i, x_{2j-1})\phi(x_{2j-1}) \\ &\quad + 2 \sum_{j=1}^{\frac{n}{2}-1} k(x_i, x_{2j})\phi(x_{2j}) + k(x_i, x_n)\phi(x_n)]. \end{aligned} \quad (2.134)$$

Eq.(2.134) can be written in the form

$$\mu\phi_i = f_i + \frac{\lambda h_s}{3} [k_{i0}\phi_0 + 4 \sum_{j=1}^{\frac{n}{2}} k_{i,2j-1}\phi_{2j-1} + 2 \sum_{j=1}^{\frac{n}{2}-1} k_{i,2j}\phi_{2j} + k_{i,n}\phi_n] \quad (2.135)$$

where $\phi_i = \phi(x_i)$, $f_i = f(x_i)$ and $k_{i,n} = k(x_i, x_n)$.

Applying SR for solving VIE of the second kind, we get

$$\mu\phi_i = f_i + \frac{\lambda h_s}{3} [k_{i,0}\phi_0 + 4 \sum_{j=1}^{\frac{i}{2}} k_{i,2j-1}\phi_{2j-1} + 2 \sum_{j=1}^{\frac{i}{2}-1} k_{i,2j}\phi_{2j} + k_{i,i}\phi_i]. \quad (2.136)$$

2.7.2 Collocation method for FIEs and VIEs

This method depends on approximating the solution $\phi(x)$ of the FIE (??) of the second kind as a partial sum of N linearly independent functions $\psi_1, \psi_2, \psi_3, \dots, \psi_N$ on the interval $[a, b]$ s.t.

$$S_N(x) = \sum_{i=1}^N c_i \psi_i(x) \quad (2.137)$$

and getting N conditions that give us N equations required for determining the N coefficients ($c_1, c_2, \dots, c_N$) of the Appr. solution (2.137), so we get this conditions by insisting the error vanishes at N -points $x_1, x_2, \dots, x_N$ where $x_i = a + ih_c$ and $h_c = \frac{b-a}{N}$. Substituting from Eq.(2.137) into Eq.(??) yields an error $E(x, c_1, c_2, \dots, c_N)$ s.t.

$$\mu S_N(x) = f(x) + \lambda \int_a^b k(x, t) S_N(t) dt + E(x, c_1, c_2, \dots, c_N). \quad (2.138)$$

Insisting the error in Eq.(2.138) vanishes at $x = x_i$, so Eq.(2.138) becomes

$$\mu S_N(x_i) = f(x_i) + \lambda \int_a^b k(x_i, t) S_N(t) dt, \quad 1 \leq i \leq N. \quad (2.139)$$

Also, Using CM for solving VIE of the second kind yields

$$\mu S_N(x_i) = f(x_i) + \lambda \int_0^{x_i} k(x_i, t) S_N(t) dt, \quad 1 \leq i \leq N. \quad (2.140)$$

2.7.3 Galerkin method for FIEs and VIEs

This method depends on approximating the solution $\phi(x)$ of the FIE (??) of the second kind as Eq.(2.137) and makes the error orthogonal to N linearly independent functions $\psi_1(x)$, $\psi_2(x)$, $\psi_3(x)$, ..., $\psi_N(x)$ on the interval $[a, b]$.

Substituting from Eq.(2.137) into Eq.(??) yields $E(x, c_1, c_2, \dots, c_N)$ s.t.

$$E(x, c_1, c_2, \dots, c_N) = \mu S_N(x) - f(x) - \lambda \int_a^b k(x, t) S_N(t) dt. \quad (2.141)$$

From GM, the error in Eq.(2.141) is orthogonal to N linearly independent functions $\chi_1(x)$, $\chi_2(x)$, $\chi_3(x)$, ..., $\chi_N(x)$ on the interval $[a, b]$, so we have

$$\int_a^b \chi_j(y) E(y, c_1, c_2, \dots, c_N) dy = 0. \quad (2.142)$$

Substituting from Eq.(2.141) in Eq.(2.142) yields

$$\int_a^b \chi_j(y) \left[\mu S_N(y) - f(y) - \int_a^b k(y, t) S_N(t) dt \right] dy = 0, \quad 1 \leq j \leq N. \quad (2.143)$$

Eq.(2.143) can be written in the form

$$\int_a^b \chi_j(y) \left[\mu S_N(y) - \int_a^b k(y, t) S_N(t) dt \right] dy = \int_a^b \chi_j(y) f(y) dy, \quad (2.144)$$

$1 \leq j \leq N.$

Substituting from (2.137) in Eq.(2.144) yields

$$\int_a^b \chi_j(y) \left[\mu \sum_{i=1}^N c_i \psi_i(y) - \int_a^b k(y, t) \sum_{i=1}^N c_i \psi_i(t) dt \right] dy = \int_a^b \chi_j(y) f(y) dy, \quad (2.145)$$

$1 \leq j \leq N.$

Applying GM for solving VIE of the second kind yields

$$\begin{aligned} \int_a^b \chi_j(y) \left[\mu \sum_{i=1}^N c_i \psi_i(y) - \int_a^x \left(k(y, t) \sum_{i=1}^N c_i \psi_i(t) \right) dt \right] dy \\ = \int_a^b \chi_j(y) f(y) dy, \quad 1 \leq j \leq N. \end{aligned} \quad (2.146)$$

2.7.4 Homotopy Analysis Method

Consider the following equation

$$\mathcal{N}(\phi(x, t)) = 0, \quad x \in [a, b], \quad t \in [0, T] \quad (2.147)$$

where $\mathcal{N}$ denotes the (linear or nonlinear) operator and $\phi(x, t)$ is an unknown function. Firstly, we describe the homotopy operator $\mathcal{H}$ as

$$\mathcal{H}(\Phi, p) = (1 - p) (\Phi(x, t; p) - \phi_0(x, t)) - ph\mathcal{N}(\Phi(x, t; p)), \quad (2.148)$$

where $p \in [0, 1]$ is the embedding parameter, $h \neq 0$ describes the convergence control parameter and $\phi_0(x, t)$ denotes the initial approximate solution of (2.147).

Suppose that $\mathcal{H}(\Phi, p) = 0$, then we obtain the zero order deformation equation

$$(1 - p) (\Phi(x, t; p) - \phi_0(x, t)) = ph\mathcal{N}(\Phi(x, t; p)). \quad (2.149)$$

If we put $p = 0$, we get $\Phi(x, t; 0) - \phi_0(x, t) = 0$, which indicates that $\Phi(x, t; 0) = \phi_0(x, t)$. For $p = 1$, we have $\mathcal{N}(\Phi(x, t; 1)) = 0$, which implies that $\Phi(x, t; 1) = \phi(x, t)$, where $\phi(x, t)$ is the solution of (2.147). Thus, the variety of parameter $p : 0 \rightarrow 1$ corresponds with the transformation of problem from the trivial to the original problem (and with the variation of solution from $\phi_0(x, t)$ to $\phi(x, t)$). If we get the Maclaurin expansion of the function $\Phi(x, t; p)$ with respect to p , we have

$$\Phi(x, t; p) = \Phi(x, t; 0) + \sum_{m=1}^{\infty} \frac{1}{m!} \frac{\partial^m \Phi(x, t; p)}{\partial p^m} \Big|_{p=0} p^m. \quad (2.150)$$

By denoting

$$\psi_m(x, t) = \frac{1}{m!} \frac{\partial^m \Phi(x, t; p)}{\partial p^m} \Big|_{p=0}, \quad m = 0, 1, 2, 3, \dots, \quad (2.151)$$

Eq.(2.150) becomes

$$\begin{aligned} \Phi(x, t; p) &= \psi_0(x, t) + \sum_{m=1}^{\infty} \psi_m(x, t) p^m \\ &= \sum_{m=0}^{\infty} \psi_m(x, t) p^m. \end{aligned} \quad (2.152)$$

If the series in (2.152) is convergent at $p = 1$, then we obtain

$$\phi(x, t) = \sum_{m=0}^{\infty} \psi_m(x, t). \quad (2.153)$$

Now, to get the function $\psi_m(x, t)$, we differentiate Eq.(2.149) m times with respect to the parameter p , divide the received result by $m!$ and substitute $p = 0$. Thus, we get m^{th} order deformation equation

$$\psi_m(x, t) - \chi_m \psi_{m-1}(x, t) = h R_m(\bar{\psi}_{m-1}, x, t), \quad m > 0, \quad (2.154)$$

where $\bar{\psi}_{m-1} = \{\psi_0(x, t), \psi_1(x, t), \dots, \psi_{m-1}(x, t)\}$,

$$\chi_m = \begin{cases} 0 & m \leq 1 \\ 1 & m > 1 \end{cases} \quad (2.155)$$

and

$$R_m(\bar{\psi}_{m-1}, x, t) = \frac{1}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial p^{m-1}} \mathcal{N} \left(\sum_{i=0}^{\infty} \psi_i(x, t) p^i \right) \right) \Big|_{p=0}. \quad (2.156)$$

Therefore, the approximate solution is given by

$$\hat{\phi}_M(x, t, h) = \sum_{m=0}^M \psi_m(x, t), \quad M = 1, 2, \dots \quad (2.157)$$

2.7.5 Homotopy Perturbation Method

We describe the homotopy perturbation method as [25, 42, 62] for a general type of the nonlinear differential equation with boundary conditions

$$A(u) - f(r) = 0, \quad r \in \Omega, \quad (2.158)$$

$$B\left(u, \frac{\partial u}{\partial n}\right) = 0, \quad r \in \Gamma, \quad (2.159)$$

where A is a general differential operator, B is a boundary operator, $f(r)$ is an known analytical function and Γ is the boundary of the domain. The operator A can be divided into two parts L and N where L is a linear operator and N is a nonlinear operator. Therefore, equation (2.158) can be rewritten as

$$L(u) + N(u) - f(r) = 0. \quad (2.160)$$

By the homotopy technique, we define a homotopy function $H(r, p) : \Omega \times [0, 1] \rightarrow R$ as

$$H(u, p) = (1 - p) [L(u) - L(u_0)] + p[A(u) - f(r)] = 0, \quad p \in [0, 1], r \in \Omega \quad (2.161)$$

where $p \in [0, 1]$ and u_0 is an initial approximation for (2.158) with

$$H(u, 0) = L(u) - L(u_0) = 0, \quad H(u, 1) = A(u) - f(r) = 0. \quad (2.162)$$

We assume that the solution of (2.158) can be written as a power series in p

$$v = \sum_{k=0}^{\infty} p^k u_k \quad (2.163)$$

Substituting from (2.163) in (2.161) and comparing the coefficients of powers of p yields a successive procedure to determine u_k . Finally, by setting $p = 1$, we obtain the solution of (2.158).

For applying Laplace homotopy perturbation method we have steps the first we applied Laplace transform on equation and then applied Laplace homotopy perturbation method. Finally, we applied Laplace inverse to get the solution numerically.

Comparison between the homotopy analysis method and homotopy perturbation method [81]

Both methods “Homotopy Perturbation Method and Homotopy Analysis Method” are in principle based on Taylor series with respect to an embedding parameter. Besides, both can give very good approximations by means of a few terms, if initial guess and auxiliary linear operator are good enough. The difference is that, “the homotopy perturbation method” had to use a good enough initial guess, but this is not absolutely necessary for the homotopy analysis method. This is mainly because the homotopy analysis method contains the auxiliary parameter(h), which provides us with a simple way to adjust and control the convergence region and rate of solution series. So, the homotopy analysis method is more general.

2.7.6 Adomian Decomposition Method

In this section, we illustrate the idea of Adomian decomposition method as [91, 17]. Let us consider the nonlinear differential equation is

$$L(u) + R(u) + N(u) - g(r) = 0, \quad (2.164)$$

where L is the highest derivative order in time which have inverse operator L^{-1} , R is a linear differentiable operator and N is a nonlinear differentiable operator. Applying the inverse operator L^{-1} to both sides of (2.164) and using the given condition

$$u = f - L^{-1}[R(u) + N(u)], \quad (2.165)$$

where the function $f(x)$ represents the terms arising from integration the source term $g(x)$.

According to Adomian decomposition method the approximate solution $u(x)$ is defined by the series:

$$u(x) = \sum_{n=0}^N u_n(x), \quad N \in \mathcal{N}, \quad (2.166)$$

where the components $u_n(x)$ are determined from the following relations

$$u_0(x) = f(x). \quad (2.167)$$

$$u_{k+1}(x) = -L^{-1} [R(u_k(x)) + N(u_k(x))], \quad k \geq 0. \quad (2.168)$$

The nonlinear operator $N(u)$ can be decomposed into an infinite series of a polynomials as

$$N(u) = \sum_{k=0}^{\infty} A_k. \quad (2.169)$$

Where A_k are so called the Adomian polynomials which given by

$$A_k = \frac{1}{k!} \left[\frac{d^k}{d\lambda^k} \left\{ N \left(\sum_{i=0}^{\infty} \lambda^i u_i(x) \right) \right\} \right]_{\lambda=0}. \quad (2.170)$$

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